

Graded Voxels in Three Dimensions

Galerkin Cubes with a Polynomial Field and Contrast: the Source of their Error, its Cure, and Closed-Form Moment Integrals

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Abstract

Voxel schemes represent a heterogeneous elastic medium as cells of constant properties, and in a smoothly varying medium that representation caps them at second order in the cell size whatever the cell carries. We build Galerkin cubes that carry polynomial moments of the nine-component state, displacement and an independent strain, with a contrast that is itself a polynomial in the cell: first the cube with a linear field and a linear contrast, then the one with both quadratic. On a sphere whose contrast falls smoothly to zero at its surface, against its exact solution, the linear cube's discretisation converges at fourth order (apparent orders 4.01 and 4.13) where uniform-field voxels converge at second, and at full contrast its error is 42 to 66 times below the collocation voxel at the same cell size. Its full-contrast order approaches four slowly, and we identify why. The error of the multiple-scattering term is static, and it equals, to within 6% on every grid, profile and shell width tried, the defect of the L^2 projection of the contrast profile onto the cell's field functions: the once-scattered field has a local part proportional to the contrast, and the cell can hold only its projection. Raising the degree of the contrast alone gains 8%; raising the degree of the field as well removes the defect at that order. The quadratic cube is more accurate than the linear one by factors of 4.1, 6.0, 10.0, 15.6 and 20.8 at 4 to 12 cells across, where the ratio of the two defects, computed from the profile alone, is 4.4, 6.2, 9.8, 15.6 and 20.6; it reaches with 408 cells the accuracy the linear cube reaches with 2,776. The coupling blocks of the self cell and its neighbours are obtained in closed form: a power series in the wavenumber whose coefficients are universal numbers, each a finite sum of three elementary master integrals reduced from volume to faces to edges, derived here in full and checked against the cube's Coulomb energy, the solid angle of the octant, an exact subdivision identity and independent quadrature. The far-field extraction must treat the engineering shear stress of the source consistently with the coupling kernel; a factor of two there is invisible at Born order along a cell axis.

1 Introduction

A voxel scheme for elastic scattering replaces a heterogeneous body by cubic cells, each carrying a single-site T -matrix and coupled through the background Green's tensor. In a layer of such cells the scheme converges at second order when each cell carries a uniform internal field, and a cell that also carries the first moment of its field converges at fourth order [1]. That paper also

showed that where the medium varies within a cell, a contrast sampled as constant caps every voxel at second order, and that the rule for a cell whose field has Legendre degree p and whose contrast has degree r is $\min(2p + 2, 2r + 2)$. All of that is one-dimensional, or three-dimensional only for a laterally uniform layer, where the lateral moments are never excited.

This paper builds the three-dimensional graded cell and asks what it achieves on a body with genuine three-dimensional structure: a sphere whose contrast falls smoothly to zero at its surface, which has an exact solution and no staircase. It has three results. The cell with a linear field and a linear contrast ($p = r = 1$) keeps the fourth order of its discretisation. Its error at full contrast is traced to a single source, the defect of the projection of the contrast onto the cell's field functions, and that source is removed by the cell with a quadratic field and contrast ($p = r = 2$), so that far fewer cells give the same accuracy. And the singular integrals on which both cells rest are obtained in closed form. The cell and its single-site T -matrix are described in §2 and §3, the coupling between cells in §4, the closed-form moments in §5, and the sphere in §6.

2 The graded voxel

A voxel is a cube $V = [-h, h]^3$ of half-width h centred at $\mathbf{x}_c$, with local coordinates $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_c)/h \in [-1, 1]^3$ in the axis order (z, x, y) . The field in it is the nine-component state $\boldsymbol{\psi} = (\mathbf{u}, \boldsymbol{\varepsilon})$: the displacement $\mathbf{u}$ and the engineering strain $\boldsymbol{\varepsilon}$ in Voigt order $(zz, xx, yy, 2xy, 2zy, 2zx)$. The strain is an *independent* unknown, not the gradient of $\mathbf{u}$. A basis whose strain is derived from its displacement leaves a spurious source at every face between cells, because each cell's projected displacement jumps across the face; in a layer that scheme converges at second or third order however rich the basis [1]. The quadratic 27-mode cube T -matrix is of that kind and is not considered further.

The state is expanded in the Legendre polynomials of degree at most one,

$$\boldsymbol{\psi}(\mathbf{x}) = \sum_{a=0}^3 \boldsymbol{\psi}_a L_a(\boldsymbol{\xi}), \quad L_0 = 1, \quad L_i = \xi_{i-1} \quad (i = 1, 2, 3), \quad (1)$$

36 weights per cell. The contrast operator $\boldsymbol{\Delta} = \text{diag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathbf{C})$, with ω the angular frequency, $\Delta \rho$ the density contrast and $\Delta \mathbf{C}$ the 6×6 Voigt stiffness contrast, is linear in the cell,

$$\boldsymbol{\Delta}(\boldsymbol{\xi}) = \boldsymbol{\Delta}_0 + \sum_{i=1}^3 \boldsymbol{\Delta}_i \xi_{i-1}, \quad (2)$$

its coefficients the L^2 projection of the medium onto $\{1, \xi_i\}$. The source density $\boldsymbol{\Delta}\boldsymbol{\psi}$ is then a polynomial of degree at most two. Writing m_c ($c = 0, \dots, 9$) for the ten monomials of degree at most two,

$$\boldsymbol{\Delta}(\boldsymbol{\xi}) L_b(\boldsymbol{\xi}) = \sum_{c=0}^9 \mathbf{E}_{cb} m_c(\boldsymbol{\xi}), \quad (3)$$

where $\mathbf{E}_{cb}$ collects the 9×9 coefficients: the Legendre linearisation of the one-dimensional construction [1], with no new kernel object. The Lippmann–Schwinger equation $\boldsymbol{\psi} = \boldsymbol{\psi}^0 + \int \mathbf{P}(\mathbf{x} -$

$\mathbf{x}') \Delta \psi d^3 x'$, with ψ^0 the incident state and $\mathbf{P}$ the 9×9 point propagator $[[\mathbf{G}, \mathbf{C}], [\mathbf{H}, \mathbf{S}]]$ built from the elastodynamic Green's tensor and its derivatives, is tested with each L_a over each cell (Galerkin). Cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\xi) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\xi') d^3 x d^3 x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (4)$$

and the discrete system is

$$\sum_b \mathbf{M}_{ab} \psi_{m,b} - \sum_n \sum_{b,c} \mathbf{K}_{ac}(\mathbf{R}_{mn}) \mathbf{E}_{cb}^{(n)} \psi_{n,b} = \langle L_a, \psi^0 \rangle_m, \quad (5)$$

with $\mathbf{M}_{ab} = \langle L_a, L_b \rangle \mathbf{I}_9$ the Gram matrix, $\text{diag}(8h^3, 8h^3/3, 8h^3/3, 8h^3/3)$.

Higher degree. Nothing in the construction is particular to degree one. The ten polynomials

$$Q_a \in \{1, \xi_i, \frac{1}{2}(3\xi_i^2 - 1), \xi_i \xi_j \ (i < j)\} \quad (6)$$

are orthogonal on the cell, with $\langle Q_a, Q_a \rangle = h^3 (8, \frac{8}{3}, \frac{8}{5}, \frac{8}{9})$ by kind, and their first four are the L_a . A contrast quadratic in the cell is the L^2 projection of the medium onto all ten ($r = 2$); a field quadratic in the cell is expanded in all ten ($p = 2$), 90 weights per cell. The products $Q_a Q_b$ are expanded in monomials as in Eq. (3): 10, 20 or 35 of them for a source of degree two, three or four. The blocks (4) are computed with test monomials and recombined into the Q_a , the first four rows and the first ten columns being those of the linear cell. The incident moments are $\langle Q_a, e^{i\mathbf{k} \cdot \mathbf{x}} \rangle = h^3 e^{i\mathbf{k} \cdot \mathbf{x}_c} \prod_i 2 i^{n_i} j_{n_i}(k_i h)$, with n_i the Legendre degree of Q_a on axis i and j_n the spherical Bessel function.

3 The single site

For one isolated cell the system closes on the self block $\mathbf{K}(0)$. With the source moments $\sigma_a = \langle L_a, \Delta \psi \rangle = \sum_b \mathbf{F}_{ab} \psi_b$, $\mathbf{F}_{ab} = \sum_c \langle L_a, m_c \rangle \mathbf{E}_{cb}$, the graded single site is

$$\mathbf{T}_{36} = \mathbf{F} (\mathbf{M} - \mathbf{K}(0) \mathbf{E})^{-1} \mathbf{M}, \quad (7)$$

which maps the incident field's Legendre coefficients to the source moments: force and stress means and their first moments. The first moment of the force, which a nine-component composite cannot hold, is part of the object.

Parity. Under inversion $\mathbf{x} \rightarrow -\mathbf{x}$ the displacement is odd, the strain even and ξ_i odd. With $g = 0$ (no gradient) $\mathbf{T}_{36}$ therefore splits exactly into a 15-dimensional set (strain $\times 1$, displacement $\times \xi$) and a 21-dimensional set (displacement $\times 1$, strain $\times \xi$); the coupling between them is 5×10^{-18} of the largest entry. A gradient is odd about the centre and couples the two sets linearly: $0.325 gh$ of the largest entry, for a gradient g along one axis, at $gh = 0.05, 0.1$ and 0.2 . An axis-aligned gradient leaves the symmetry of rotations by a quarter turn about its axis, which $\mathbf{T}_{36}$ satisfies to

10^{-11} .

Reciprocity. The propagator satisfies $\mathbf{W}\mathbf{P}(\mathbf{r})$ symmetric at every $\mathbf{r}$, with $\mathbf{W} = \text{diag}(\mathbf{I}_6, \frac{1}{2}\mathbf{I}_3)$, and $\mathbf{P}(-\mathbf{r}) = \mathbf{\Pi}\mathbf{P}(\mathbf{r})\mathbf{\Pi}$ with $\mathbf{\Pi} = \text{diag}(-\mathbf{I}_3, \mathbf{I}_6)$. Every block between test functions is therefore $\mathbf{W}$ -symmetric, $(\mathbf{W}\mathbf{K}_{ab})^\top = \mathbf{W}\mathbf{K}_{ab}$, and the self block obeys $\mathbf{K}_{ba}(0) = \mathbf{\Pi}\mathbf{K}_{ab}(0)\mathbf{\Pi}$ (exchange the two points of the cell). Both hold to 10^{-11} .

The Born limit. Without a gradient $\mathbf{T}_{36}$ is exact at first order in the contrast for any incident field: the source moments are $\Delta_0\langle L_a, \psi^0 \rangle$. With a gradient the first-order source $\langle L_a, (\mathbf{g} \cdot \boldsymbol{\xi})\psi^0 \rangle$ needs the incident field's *quadratic* moments, which the degree-one input drops. The error is $O(\Delta gh(kh)^2)$, with k the incident wavenumber: measured slopes 2.0001 and 2.0004 for $kh = 0.025 \rightarrow 0.05 \rightarrow 0.1$, where it reaches 8.9×10^{-5} at $gh = 0.2$ (these and the parity numbers: `scripts/measure_graded_voxel_site.py`). It is the single site's view of the rule $\min(2p+2, 2r+2)$ for a field of degree p and a contrast of degree r [1].

An isolated cube has an error floor that refinement in k does not remove. In the static limit a cube of contrast $\Delta\mathbf{C}$ under a uniform remote strain has the internal strain $\boldsymbol{\varepsilon}(\mathbf{x}) = \boldsymbol{\varepsilon}^0 + \int_V \partial\partial\mathbf{G}(\mathbf{x} - \mathbf{x}') \Delta\mathbf{C} \boldsymbol{\varepsilon}(\mathbf{x}') d^3x'$. The self-interaction $\mathbf{S}(\mathbf{x}) = \int_V \partial\partial\mathbf{G}(\mathbf{x} - \mathbf{x}') d^3x'$ is constant inside an ellipsoid and only there [2]; in a cube it concentrates towards the edges and corners. A closure that keeps averages is exact through second order in the contrast, since the second-order source needs only $\int_V \mathbf{S}$, and first errs at third order, by $\Delta\mathbf{C}^3 V \text{Var}(\mathbf{S})$; a collocation closure, which uses $\mathbf{S}(0)$, errs already at second order. Neither error depends on k . It is not what a tiled medium sees: between identical cells the faces are not physical, the exact field is smooth there, and the concentration does not arise. The order of convergence is a property of the tiled medium, the closure error a property of the isolated cell; this paper measures the former (§6).

4 The coupling between cells

The point propagator. $\mathbf{P}$ is built from $\mathbf{G}_{ij} = \frac{1}{4\pi\mu} [\delta_{ij}g_S(r) + k_S^{-2}\partial_i\partial_j(g_S - g_P)(r)]$, $g_k = e^{ikr}/r$, with μ the background shear modulus and k_P, k_S the P and S wavenumbers. Every derivative of a radial function f is a combination of $F_q = (r^{-1}d/dr)^q f$. Near the origin the closed form cancels catastrophically (the k_S^{-2} difference), so for $k_S r \leq \frac{1}{2}$ the F_q are summed term by term from the power series of g_k . This kernel equals the package's closed form to 10^{-11} where both are accurate, and its static part is the Kelvin tensor to 10^{-13} . Its static (Kelvin) part is the $t = 0$ term of the δ_{ij} series and the $t = 2$ term of the derivative series, $\frac{1}{4\pi\mu} [\delta_{ij}r^{-1} + b_2\partial_i\partial_j r]$ with $b_2 = -\frac{1}{2}(1 - \beta^2/\alpha^2)$; the dynamic remainder is at most $1/r$.

The s-form. With $\mathbf{x} = \mathbf{x}_m + \mathbf{u}$ and $\mathbf{x}' = \mathbf{x}_n + \mathbf{u}'$ the kernel sees $\mathbf{R} + \mathbf{s}$, $\mathbf{s} = \mathbf{u} - \mathbf{u}' \in [-2h, 2h]^3$, and Eq. (4) becomes

$$\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s, \quad W_{ac}(\mathbf{s}) = \prod_i w_{e_{a,i}e_{c,i}}(s_i), \quad (8)$$

with the one-dimensional cross-correlation $w_{\alpha\beta}(s) = h \int v^\alpha (v - s/h)^\beta dv$ over $v \in [\max(-1, s/h - 1), \min(1, s/h + 1)]$: continuous, a polynomial on $[-2h, 0]$ and on $[0, 2h]$, zero at $\pm 2h$, with kinks at 0 and $\pm 2h$. For cells that do not touch the kernel is smooth on each of the eight pieces and a three-dimensional Gauss rule suffices ($8n^3$ kernel evaluations against n^6 for the double integral). It agrees with six-dimensional Gauss quadrature of Eq. (4) to 10^{-9} one cell beyond contact.

The self cell and its 26 neighbours. The static part is a sum of terms $\partial^l r^m$ ($m = -1$ with $|\iota| \leq 2$, $m = 1$ with $|\iota| \leq 4$), distributions at the origin. Moving $k = \max(0, |\iota| - (m + 1)) \leq 2$ derivatives onto W leaves a kernel at most $1/r$: $\int W \partial^l f = (-1)^k \int (\partial^{\text{moved}} W) \partial^{\text{rest}} f$. Two derivatives on one axis turn the kinks of w into plane deltas $J_b \delta(s - bh)$, $b \in \{-2, 0, 2\}$, J_b the jump of w' at b . Every piece is then a box or a rectangle whose corners lie on the breakpoints, so the singular point $\mathbf{s} = -\mathbf{R}$ is either a vertex, integrated with Duffy's pyramids, or at distance $2h$ or more. The Eshelby delta is not added by hand; it is carried by the plane deltas. The construction reproduces the Eshelby sum rule $\sum_p \iint_{V \times V} \partial_p \partial_p r^{-1} = -4\pi V$ and the cube's Coulomb self-energy

$$\iint_{V \times V} \frac{d^3x d^3x'}{|\mathbf{x} - \mathbf{x}'|} = 2 \left[\frac{1}{5} (1 + \sqrt{2} - 2\sqrt{3}) - \frac{\pi}{3} + \ln((1 + \sqrt{2})(2 + \sqrt{3})) \right] (2h)^5 \quad (9)$$

to 10^{-11} , converges in the Gauss order to 10^{-11} , and satisfies the reciprocity and exchange laws of §3. The biharmonic rule $\sum_{p,q} \partial_p \partial_p \partial_q \partial_q r = -8\pi\delta$ holds to the same accuracy, which checks the delta carried by the r terms. And every block equals the sum of the 64 blocks between the two cells' half-size sub-cells, with the polynomials re-expressed on the sub-cells, to 2×10^{-15} for all (a, c) at the self cell, its touching neighbours and beyond (`scripts/check_graded_voxel_subdivision.py`).

The pieces are integrated here by quadrature; §5 evaluates the same pieces in closed form.

An independent construction. Integration by parts can move the derivatives onto the cells' polynomials instead of onto W : one derivative onto each cell, each move either differentiating the polynomial or producing a face term, which leaves $1/r$ kernels over cell×cell, face×cell and face×face, all absolutely convergent. Built in Mathematica on its own (`GradedVoxel_WeakKernel.wl`), this reproduces the sum rule and the Coulomb energy, and, as the source cell shrinks to a point, the distributional single average of the moment engine [1]; all to 10^{-14} or better.

5 The moment integrals in closed form

The blocks of the self cell and of its 26 neighbours need no quadrature. They are a power series in the wavenumber whose coefficients are pure numbers, each a finite sum of elementary master integrals. This section derives them from the definition (4); nothing is taken from elsewhere.

The series in the wavenumber. With $g_k = e^{ikr}/r = \sum_{t \geq 0} (ik)^t r^{t-1}/t!$, the Green's tensor is a sum over powers r^m , $m = t - 1 \geq -1$, of two families,

$$\mathbf{G}_{ij} = \frac{1}{4\pi\mu} \sum_{m \geq -1} \left[c_1(m) \delta_{ij} r^m + c_2(m) \partial_i \partial_j r^m \right], \quad c_1 = \frac{(ik_S)^{m+1}}{(m+1)!}, \quad c_2 = \frac{i^{m+1} (k_S^{m+1} - k_P^{m+1})}{k_S^2 (m+1)!}. \quad (10)$$

The static (Kelvin) tensor is c_1 at $m = -1$ and c_2 at $m = 1$, where $c_2 = b_2$. The propagator $\mathbf{P}$ adds up to two derivatives to $\mathbf{G}$, so every entry of every block is a sum of the scalar moments

$$\int_{V_m} \int_{V_n} L_a \partial^t r^m m_c d^3 x d^3 x' = h^{6+m-|\iota|} U_{ac}(m, \iota; \mathbf{o}), \quad |\iota| \leq 4, \quad (11)$$

with $\mathbf{o} = \mathbf{R}/2h$ the integer offset between the cells. The U are numbers: they depend on neither the cell size, the frequency nor the medium. The medium enters a block only through c_1 , c_2 and two fixed tables of 9×9 integer matrices, one for each family, that say which entry of $\mathbf{P}$ takes which derivative ∂^t . The series is that of the exponential and converges for every k ; for touching cells its term t is bounded by $(4\sqrt{3}k_S h)^t/t!$.

Even and odd powers. For even m , r^m is a polynomial and (11) is an integral of polynomials over the pieces of W , exact by a Gauss rule of sufficient order. For odd m the kernel is singular at $\mathbf{x} = \mathbf{x}'$, and for $m = \pm 1$ with enough derivatives it is a distribution. Of the derivatives in ι ,

$$k = \max\left(0, |\iota| - \min\left(m+1, \frac{m+3}{2}\right)\right) \leq 2 \quad (12)$$

are moved onto W as in §4. What remains, $\partial^{\text{rest}} r^m$, is then at most $1/r$ singular, and it is a sum of monomials times odd powers of the distance no lower than r^{-3} , by

$$\partial_i (t^\alpha |t|^n) = \alpha_i t^{\alpha-e_i} |t|^n + n t^{\alpha+e_i} |t|^{n-2} : \quad (13)$$

$1/r$ for $m = -1$, and $\delta_{ij}/r - t_i t_j/r^3$ for $m = 1$.

From a piece of W to boxes at the singular point. Let $\mathbf{t} = (\mathbf{R} + \mathbf{s})/h$ be the separation in units of h , measured from the singular point. Each piece of the differentiated W is a polynomial in $\mathbf{t}$ on a box whose corners have coordinates in $\{0, \pm 2, \pm 4\}$. On each axis an integral from l to u is the difference of two integrals from the origin, and an integral from the origin to a negative limit is, by $t \rightarrow -t$, $\pm$ one to a positive limit, the sign being the parity of the total power of t on that axis. A piece is therefore a signed sum of integrals over boxes $[0, a] \times [0, b] \times [0, c]$ with one corner at the singular point and sides 2 or 4. A plane delta of W'' fixes one coordinate at $t_i = x_0 \in \{0, \pm 2, \pm 4\}$ and leaves an integral over a rectangle at distance $|x_0|$ from the singular

point. Three master integrals remain:

$$I[p, q, r; a, b, c; m] = \int_0^a \int_0^b \int_0^c x^p y^q z^r R^m dz dy dx, \quad R^2 = x^2 + y^2 + z^2, \quad (14)$$

$$J[q, r; a; b, c; m] = \int_0^b \int_0^c y^q z^r S^m dz dy, \quad S^2 = a^2 + y^2 + z^2, \quad (15)$$

$$L[r; A; c; m] = \int_0^c z^r (A^2 + z^2)^{m/2} dz, \quad (16)$$

with m odd.

Volume to faces. The integrand of I is homogeneous of degree $d = p + q + r + m$, so $\nabla \cdot (\mathbf{x}f) = (d + 3)f$. The divergence theorem on the box, whose faces through the origin carry no flux, gives

$$(d + 3)I[p, q, r; a, b, c; m] = a^{p+1}J[q, r; a; b, c; m] + b^{q+1}J[p, r; b; a, c; m] + c^{r+1}J[p, q; c; a, b; m]. \quad (17)$$

Convergence at the origin is $d > -3$, so the factor never vanishes.

Faces to edges. Integrating $\partial_y(y^{q-1}S^{m+2}) = (q-1)y^{q-2}S^{m+2} + (m+2)y^qS^m$ over the face,

$$(m+2)J[q, r; m] = b^{q-1}L[r; \sqrt{a^2 + b^2}; c; m+2] - \delta_{q1}L[r; a; c; m+2] - (q-1)J[q-2, r; m+2], \quad (18)$$

which lowers q two at a time, and r likewise, to $q, r \in \{0, 1\}$; at $q = 1$ the last term is absent and J is a single edge term. For $q = r = 0$, writing $S^m = S^{m-2}(a^2 + y^2 + z^2)$ and using the same identity for the two quadratic terms,

$$(m+2)J[0, 0; m] = m a^2 J[0, 0; m-2] + b L[0; \sqrt{a^2 + b^2}; c; m] + c L[0; \sqrt{a^2 + c^2}; b; m], \quad (19)$$

which climbs from the one transcendental base case

$$J[0, 0; a; b, c; -3] = \frac{1}{a} \arctan \frac{bc}{a\sqrt{a^2 + b^2 + c^2}}, \quad (20)$$

$1/a$ times the solid angle the rectangle subtends at the origin. On a face through the singular point, $a = 0$, the integrand is homogeneous in (y, z) and the plane form of (17) gives $(q + r + m + 2)J = b^{q+1}L[r; b; c; m] + c^{r+1}L[q; c; b; m]$.

Edges. Integrating $\partial_z(z^{r-1}(A^2 + z^2)^{(m+2)/2})$,

$$(m+2)L[r; m] = c^{r-1}(A^2 + c^2)^{(m+2)/2} - \delta_{r1}A^{m+2} - (r-1)L[r-2; m+2], \quad (21)$$

and for $r = 0$, from $\partial_z(z(A^2 + z^2)^{m/2})$,

$$(m+1)L[0; m] = c(A^2 + c^2)^{m/2} + m A^2 L[0; m-2], \quad L[0; -1] = \operatorname{arsinh} \frac{c}{A}, \quad L[0; -3] = \frac{c}{A^2 \sqrt{A^2 + c^2}}. \quad (22)$$

For $A = 0$, $L[r; m] = c^{r+m+1}/(r+m+1)$. Every moment is thus a rational combination of square roots of integers, one inverse hyperbolic sine and one arctangent per box; the divisors are small odd integers, and the recursions are evaluated at forty digits.

Checks. Expanding $\prod_i(2-s_i)$ on the octant, the cube’s Coulomb self-energy (9) at $h = 1$ is $8 \sum_{p,q,r \in \{0,1\}} 2^{3-p-q-r} (-1)^{p+q+r} I[p, q, r; 2, 2, 2; -1]$; the master integrals reproduce its closed form to 25 digits. The flux of $\mathbf{x}/r^3$ through the three far faces of any box is the octant’s solid angle, and $aJ[0, 0; a; b, c; -3] + bJ[0, 0; b; a, c; -3] + cJ[0, 0; c; a, b; -3] = \pi/2$ holds to 25 digits for boxes of sides $(2, 2, 2)$, $(2, 4, 2)$ and $(4, 2, 4)$. Lines, faces and boxes agree with direct numerical integration to the accuracy of that integration, and with direct symbolic integration, one variable at a time, to 25 digits wherever it terminates (18 of 21 cases tried). Assembled, every static term of the kernel agrees with the Duffy quadrature of §4 to 10^{-9} for all pairs (a, c) at the self cell and at face, edge and corner neighbours, and the complete blocks, static and dynamic, agree with the quadrature blocks to 10^{-10} at $k_s h = 0.0625$ and 0.5 for the linear field and at $k_s h = 0.25$ for the quadratic field of §2 (`cubic_scattering/graded_voxel/moments.py`, `blocks.near_block_series`). For cells that do not touch the kernel is smooth and the Gauss rule of §4 is used; the same closed forms apply there with longer boxes, and have not been tabulated.

6 The smoothly graded sphere

The body and its exact solution. A sphere of radius $a = 10$ m has a homogeneous core $r < a/2$ with the contrast $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100 \text{ kg m}^{-3}$ in a background $\alpha = 5 \text{ km s}^{-1}$, $\beta = 3 \text{ km s}^{-1}$, $\rho = 2500 \text{ kg m}^{-3}$. Across the shell $a/2 < r < a$ the contrast is multiplied by the smoothstep $10x^3 - 15x^4 + 6x^5$, $x = (a-r)/(a/2)$, so it vanishes like $(a-r)^3$ at the surface: there is no staircase. Its per-order T -matrices are exact (Takeuchi–Saito integration, two implementations agreeing to 2×10^{-11}) [1]. The incident wave is a P wave along z , at $k_s a = 0.5$ and 1.

Arms and measurement. At the same cell size, on a grid of n^3 cells across the bounding cube: the collocation uniform-field voxel with each cell at the profile’s centre value (T_9), which keeps the cells whose centre lies inside the sphere; the Galerkin uniform-field voxel ($p = r = 0$); and the graded first-moment voxel ($p = r = 1$). A Galerkin cell carries the L^2 projection of its contrast (six-point Gauss rule per axis), so the Galerkin arms keep every cell overlapping the sphere (408 cells at $n = 8$, against 280): keeping only cells whose centre lies inside drops contrast from partly filled shell cells, and the uniform Galerkin voxel then converges at order 1.07 instead of 2. The error is the largest far-field difference from the exact solution over nine angles, relative to its peak, at 5×10^8 radii; at 5×10^4 radii the near-field terms that the asymptotic far field omits are 1.4×10^{-4} , larger than the graded voxel’s own error.

The solve. A block $\mathbf{K}_{ac}(\mathbf{o})$ depends only on the offset $\mathbf{o}$ between two cells of the grid. The background is isotropic and the cube is mapped to itself by its 48 signed permutations $\mathbf{Q}$, so $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^\top$ with $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$, where $\mathbf{R}_6$ is the representation of $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^\top$ on

Voigt vectors (for a signed permutation it is the same for tensor and engineering components). With the moments of the test and source polynomials transforming as $L_a(\mathbf{Q}\boldsymbol{\eta}) = \sum_b T_{ab} L_b(\boldsymbol{\eta})$ and $m_c(\mathbf{Q}\boldsymbol{\eta}) = \sum_d U_{cd} m_d(\boldsymbol{\eta})$,

$$\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^\top, \quad (23)$$

so one block is computed per orbit of offsets and the others follow exactly (to 10^{-11} against direct computation). The coupling is then a convolution over the grid, applied by FFT on a $(2n-1)^3$ circular embedding, and the system is solved by GMRES preconditioned with each cell's own block. On the grids where both run, this solve and the dense one give far-field errors that agree to 5×10^{-9} relative (`cubic_scattering/graded_voxel/fft.py`); at $n = 16$ (99,936 unknowns) it takes 105 s and 3.9 GB.

The discretisation alone. At a contrast 10^{-6} of the above, multiple scattering is negligible and the error is that of the discretisation (Table 1). The graded voxel converges at fourth order, with apparent orders 4.01 and 4.13 on the finest pair; both uniform-field voxels are of second order. At $n = 4$ a cell is as wide as the graded shell and the orders there are not asymptotic.

n	$k_S a = 0.5$			$k_S a = 1$		
	T_9	$p = r = 0$	$p = r = 1$	T_9	$p = r = 0$	$p = r = 1$
4	1.04×10^{-1}	8.85×10^{-3}	7.53×10^{-5}	1.16×10^{-1}	3.45×10^{-2}	2.32×10^{-4}
6	4.71×10^{-3}	3.61×10^{-3}	5.03×10^{-6}	8.26×10^{-3}	1.42×10^{-2}	9.14×10^{-5}
8	2.71×10^{-3}	2.06×10^{-3}	1.59×10^{-6}	5.17×10^{-3}	8.15×10^{-3}	2.78×10^{-5}
order 6 $\rightarrow$ 8	1.92	1.96	4.01	1.63	1.93	4.13

Table 1. Weak contrast (10^{-6}): the discretisation error of the far field.

Full contrast. With multiple scattering (Table 5) the graded voxel's error falls to 6.3×10^{-5} and 7.6×10^{-5} at $n = 8$, 42 and 66 times below the collocation voxel at the same cell size; at $k_S a = 1$ it is below the collocation voxel at $n = 24$ (4.2×10^{-4} with 64,872 unknowns, against 7.6×10^{-5} with 14,688). Its apparent order between $n = 6$ and 8 is 3.35 and 3.40: below four, and rising. The part of the error that multiple scattering adds is carried by the second-order Born term T_2 (one application of the coupling). Computed on its own, which needs no solve and so reaches finer grids, its apparent order at $k_S a = 0.5$ rises through 2.85, 3.40, 3.52 and 3.65 from $n = 4$ to 12 (errors 4.1×10^{-2} to 1.1×10^{-3} of T_2 ; `scripts/measure_graded_voxel_t2.py`). At $k_S a = 1$ the sequence is almost the same: apparent orders 2.88, 3.42, 3.54, 3.67 (errors 4.2×10^{-2} to 1.1×10^{-3}). The slow approach does not depend on the frequency. The Born term contributes little: its error, 7.5×10^{-5} of T_1 at $n = 4$ and 1.3×10^{-7} at $n = 16$, is below T_2 's share of the full error at every n .

The full solve, taken to $n = 16$ (Table 4), follows the same law. Its apparent order rises through 3.54, 3.67, 3.75 and 3.80 from $n = 8$ to 16 at $k_S a = 0.5$, and through 3.56, 3.67, 3.75, 3.80 at $k_S a = 1$, within 0.02 of the apparent orders of T_2 on the same grids. The correction to fourth order is large, but its own order is not fixed by these grids: on the even grids from $n = 10$, with $h = 1/n$, $Ah^4 + Bh^5$ and $Ah^4 + Ch^6$ fit T_2 at $k_S a = 0.5$ to 0.21% and 0.23%, and odd n , for

which the sphere's centre falls at a cell centre rather than a vertex, scatter by 1 to 2% about the curve of the even ones.

What sets the approach. Three causes are separable (Table 2; `scripts/measure_graded_voxel_resolution.py`). The profile $s(x)$ is only C^2 : its third derivative jumps at $r = a/2$ and at $r = a$. Replaced by the C^∞ step $s(x) = e^{-1/x}/(e^{-1/x} + e^{-1/(1-x)})$, whose derivatives all vanish at both ends, the approach is not faster but slower (apparent orders 3.31 to 3.77, irregular), so the jumps are not its cause; that step is steeper, with a maximum slope of 2 against 1.875 and much larger higher derivatives. Widening the shell instead, with the core at $a/4$, so that $n = 16$ puts six cells across it rather than four, speeds the approach for both profiles: the apparent order reaches 3.92 with s and 3.88 with the C^∞ step. The full solve follows T_2 throughout. The approach to fourth order is therefore governed by how finely the cells resolve the variation of the contrast.

n	core $a/2$ (4 cells across the shell at $n = 16$)		core $a/4$ (6 cells)	
	C^2 step	C^∞ step	C^2 step	C^∞ step
8	6.28×10^{-5}	8.33×10^{-5}	2.91×10^{-5}	4.01×10^{-5}
10	2.85×10^{-5}	3.98×10^{-5}	1.26×10^{-5}	1.78×10^{-5}
12	1.46×10^{-5}	2.10×10^{-5}	6.25×10^{-6}	9.38×10^{-6}
14	8.18×10^{-6}	1.18×10^{-5}	3.44×10^{-6}	5.26×10^{-6}
16	4.93×10^{-6}	7.35×10^{-6}	2.04×10^{-6}	3.13×10^{-6}
order $14 \rightarrow 16$	3.80	3.52	3.92	3.88

Table 2. Full contrast, $k_s a = 0.5$: the graded voxel's far-field error for two shell profiles and two shell widths.

The error of T_2 is a projection defect. Two measurements identify it. First, it is static. At $n = 8$ the relative error of T_2 is 5.3, 5.2, 4.8, 4.9 and 5.1×10^{-3} at $k_s a = 0.125, 0.25, 0.5, 1$ and 2 , while that of the Born term grows as the fourth power of the frequency (1.6×10^{-6} , 2.8×10^{-5} and 4.5×10^{-4} at $k_s a = 0.5, 1$ and 2). Second, with Π_1 the L^2 projection onto the functions linear in each cell, it equals the defect

$$D_1 = \frac{\sum_{\text{cells}} \int (s - \Pi_1 s)^2 dV}{\sum_{\text{cells}} \int s^2 dV} \quad (24)$$

of the profile s , computed from the profile alone: the ratio of T_2 's error to D_1 lies between 1.03 and 1.06 on every grid from $n = 8$ to 16, odd and even, for both profiles and both shell widths ($D_1 = 4.6 \times 10^{-3}$ at $n = 8$ and 3.6×10^{-4} at $n = 16$ for the sphere of Table 5). The apparent order of D_1 itself between $n = 14$ and 16 is 3.79, which is the 3.80 of the full solve. The reading is this. The once-scattered field has a local part proportional to the contrast, from the singular part of the propagator. The cells hold only its projection onto their linear field functions, so the second-order term pairs the contrast with Π_1 of itself, $\int (\Pi_1 s)^2$ in place of $\int s^2$, and loses D_1 . The Born term is spared because the projection keeps each cell's mean and first moments of the contrast. This is an argument that fits the measurements, not a derivation. It makes one

prediction that was tested: the defect belongs to the field functions, not to the representation of the contrast. With the contrast quadratic in each cell (its ten orthogonal polynomials of degree two, twenty source monomials of degree three; the blocks pass the same subdivision identity) and the field still linear, the full-contrast error at $k_s a = 0.5$ falls only from 5.47×10^{-4} , 6.28×10^{-5} and 1.46×10^{-5} to 5.03×10^{-4} , 5.86×10^{-5} and 1.35×10^{-5} at $n = 4, 8$ and 12 : by 8%, where the defect of a quadratic projection is 10 and 21 times below D_1 at $n = 8$ and 12 . A field quadratic in the cell is what the argument calls for.

The cure: a field quadratic in the cell. With field and contrast both quadratic ($p = r = 2$, §2) the defect becomes D_2 , that of the projection Π_2 onto the ten polynomials of Eq. (6), and the argument predicts that the full-contrast error falls by D_1/D_2 . It does (Table 3): the measured gain over the linear cell is 4.1, 6.0, 10.0, 15.6 and 20.8 from $n = 4$ to 12 , and D_1/D_2 , computed from the profile alone, is 4.4, 6.2, 9.8, 15.6 and 20.6. The apparent order of the quadratic cell rises through 3.91, 5.08 and 5.57 and is 5.22 on the last pair; D_2 itself has apparent orders 3.73, 5.10, 5.62 and 5.19 on the same pairs, so the dip is the defect's own and not a new source of error. A defect quadratic in a projection error of third order is of sixth order, and the cell is approaching it. At $n = 12$ the error, 7.0×10^{-7} , is within an order of magnitude of the accuracy to which the comparison itself has been verified (the Born term's error stops falling near 10^{-7} at low frequency), so finer grids would need a sharper reference.

n	unknowns		error		gain	D_1/D_2	order ($p = r = 2$)
	$p = r = 1$	$p = r = 2$	$p = r = 1$	$p = r = 2$			
4	2,304	5,760	5.47×10^{-4}	1.33×10^{-4}	4.1	4.4	
6	7,488	18,720	1.65×10^{-4}	2.72×10^{-5}	6.0	6.2	3.91
8	14,688	36,720	6.28×10^{-5}	6.30×10^{-6}	10.0	9.8	5.08
10	27,648	69,120	2.85×10^{-5}	1.82×10^{-6}	15.6	15.6	5.57
12	46,080	115,200	1.46×10^{-5}	7.02×10^{-7}	20.8	20.6	5.22

Table 3. Full contrast, $k_s a = 0.5$: the cell with a linear field and contrast against the cell with both quadratic. The gain is the ratio of the two errors; D_1/D_2 is the ratio of the projection defects of the profile, Eq. (24) with Π_1 and with Π_2 .

Fewer cells. A more accurate cell is worth having because fewer are needed. The linear cell reaches 4.9×10^{-6} at $n = 16$ with 2,776 cells and 99,936 unknowns; the quadratic cell reaches 6.3×10^{-6} at $n = 8$ with 408 cells and 36,720 unknowns, and 1.8×10^{-6} at $n = 10$ with 768 cells and 69,120. The number of coupling blocks, and of offsets between cells, falls with the number of cells; the work per block rises with the size of the cell's basis.

A convention the Born limit cannot see. The shear entries of the source are engineering stress $2\sigma_{pq}$: the contrast operator maps the engineering strain γ_{pq} to $2\Delta\mu\gamma_{pq}$, and the kernel's shear columns carry a factor $\frac{1}{2}$. The far field must take the tensor stress σ_{pq} from them, half the entry. Radiating the entry itself doubles the shear stress, and a P wave along a cell axis has no shear stress at Born order, so the weak-contrast test cannot see it. It appears at second order, through the density-shear cross terms, as an error in T_2 that does not fall with refinement

n	unknowns	$k_S a = 0.5$		$k_S a = 1$	
		error	order	error	order
4	2,304	5.47×10^{-4}		6.34×10^{-4}	
6	7,488	1.65×10^{-4}	2.96	2.02×10^{-4}	2.82
8	14,688	6.28×10^{-5}	3.35	7.60×10^{-5}	3.40
10	27,648	2.85×10^{-5}	3.54	3.43×10^{-5}	3.56
12	46,080	1.46×10^{-5}	3.67	1.76×10^{-5}	3.67
14	68,832	8.18×10^{-6}	3.75	9.85×10^{-6}	3.75
16	99,936	4.93×10^{-6}	3.80	5.93×10^{-6}	3.80

Table 4. Full contrast: the graded voxel ($p = r = 1$) solved by FFT to $n = 16$; the order is the apparent order from the previous row.

n	$k_S a = 0.5$			$k_S a = 1$		
	T_9	$p = r = 0$	$p = r = 1$	T_9	$p = r = 0$	$p = r = 1$
4	1.03×10^{-1}	7.70×10^{-3}	5.47×10^{-4}	1.15×10^{-1}	3.24×10^{-2}	6.34×10^{-4}
6	4.66×10^{-3}	3.04×10^{-3}	1.65×10^{-4}	8.15×10^{-3}	1.32×10^{-2}	2.02×10^{-4}
8	2.64×10^{-3}	1.68×10^{-3}	6.28×10^{-5}	5.01×10^{-3}	7.52×10^{-3}	7.60×10^{-5}
order $6 \rightarrow 8$	1.98	2.05	3.35	1.70	1.95	3.40

Table 5. Full contrast: the far-field error of the three voxels.

(4.65% and 4.05% at $n = 4$ and 6, against 4.22% and 1.31% when the far field agrees with the kernel for every source component).

7 Discussion

What three dimensions change. Two things differ from the layer. The lateral first moments are now excited, and they couple through the near field of neighbouring cells; and the discrete source, projected cell by cell, jumps by $O(h^2)$ across every face between cells, where in the layer only the faces normal to the stratification exist. The Born term does not see either: its error is the quadrature error of the source, and it is of fourth order. The multiple-scattering term sees both, but neither costs an order: its apparent order rises towards four, and the full solve follows it (§6). The approach is slow where few cells span the variation of the contrast, and does not depend on the frequency or on the smoothness of the profile. The error of the multiple-scattering term is the defect (24) of the profile’s projection onto the cells’ field functions (§6), a static quantity three orders of magnitude above the Born term’s error at these frequencies.

Galerkin and collocation. With the lattice couplings then available, a uniform-field Galerkin cell was measured several times worse than the collocation cell [1]. Here, with every block computed exactly, the uniform-field Galerkin voxel is comparable to the collocation voxel at second order, and the graded Galerkin voxel is far better. The double average is the route to fourth order when the coupling is exact.

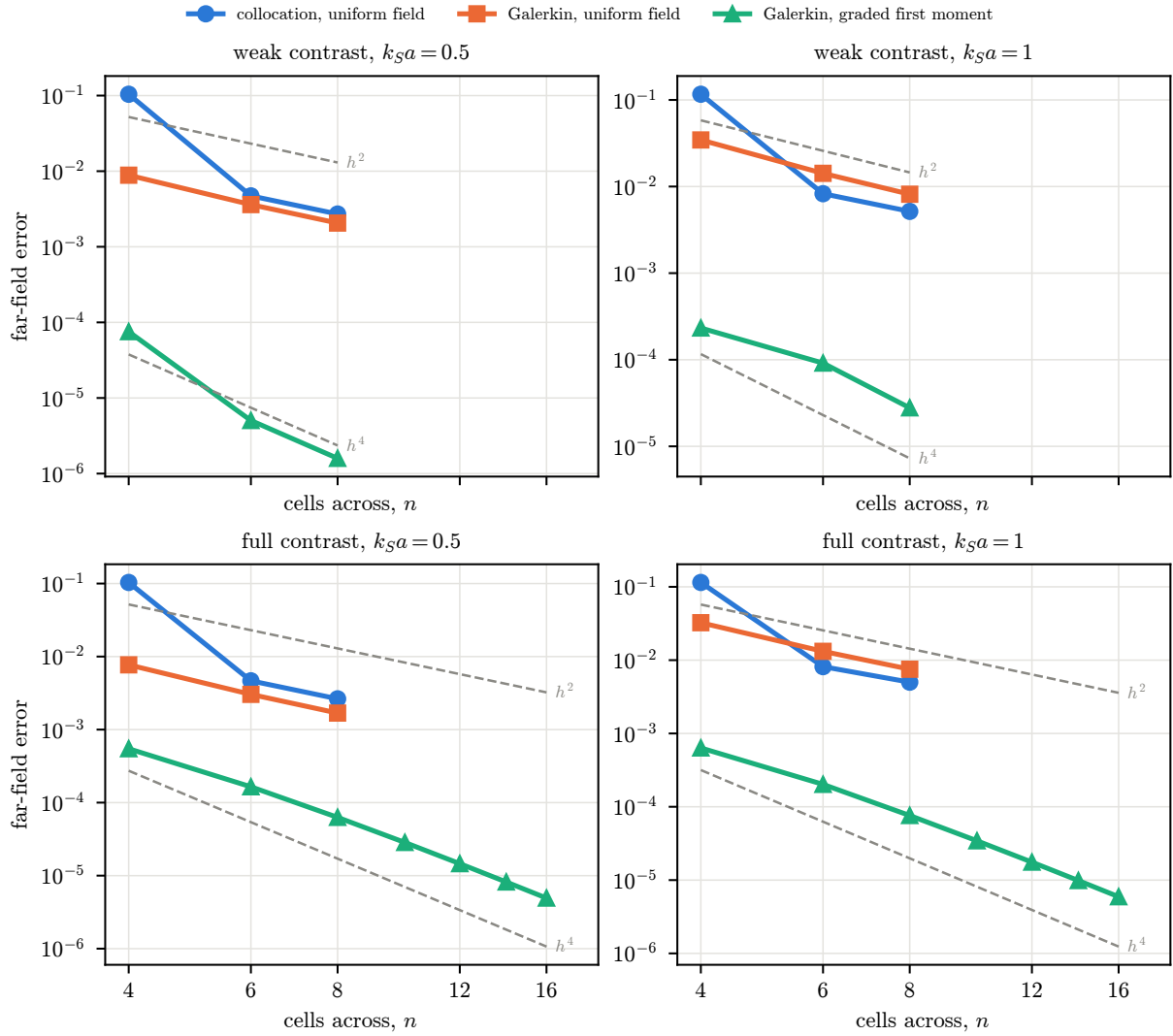


Figure 1. Far-field error against the number of cells across the sphere, for the collocation and Galerkin uniform-field voxels and the graded first-moment voxel; dashed guides show h^2 and h^4 . Top: weak contrast (the discretisation alone). Bottom: full contrast, with the graded voxel solved by FFT to $n = 16$ (Table 4).

Cost. The linear cell has 36 unknowns against 9, the quadratic cell 90. On the sphere at full contrast the linear cell reaches 7.6×10^{-5} with 14,688 unknowns at $k_{sa} = 1$; the collocation voxel reaches only 4.2×10^{-4} with 64,872. The quadratic cell reaches 6.3×10^{-6} with 36,720 unknowns where the linear cell needs 99,936 for 4.9×10^{-6} . With an FFT matrix-vector product the quadratic cell's kernel has 90×315 entries per offset against 36×90 , and at $n = 12$ the solve takes 587s and 6.8 GB on a workstation. The blocks of the self cell and its neighbours are universal constants times powers of h and k (§5) and are computed once; the blocks between cells that do not touch cost a three-dimensional Gauss rule over the eight pieces of the autocorrelation.

The representation of the medium. The order of a cell is set by how well it represents the medium, twice over: the contrast must be represented, and the field must be able to carry the variation that the contrast imprints on it. A contrast of higher degree than the field is wasted;

the measured gain from it is 8%.

8 Conclusions

A Galerkin cube that carries the mean and first moments of displacement and an independent strain, with a contrast linear in the cell, converges at fourth order in its discretisation (at Born order) on a three-dimensional body with an exact solution, where cells of constant properties converge at second. At full contrast it is between one and two orders of magnitude more accurate than a uniform-field voxel at the same cell size, and its error there has one identified source: the multiple-scattering term loses the defect of the L^2 projection of the contrast onto the cell's field functions, a static quantity that predicts the error to within 6% and that explains the slow approach of the apparent order to four. A cube whose field and contrast are both quadratic removes the defect at that order. Its measured gain over the linear cube, 4.1 to 20.8 between 4 and 12 cells across, is the ratio of the two defects, and it reaches a given accuracy with several times fewer cells. A quadratic contrast alone does not. The singular integrals of the coupling blocks are available in closed form, as a power series in the wavenumber with universal coefficients built from three elementary master integrals; they agree with the quadrature construction of the same blocks to 10^{-10} . The far field of such a scheme must take the source's engineering shear stress as the coupling kernel does; the Born limit along a cell axis cannot reveal a mismatch there.

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